

No need to submit. Please attempt all these questions yourself and reach out for any difficulties.

1. Find the $\limsup$, $\liminf$ and $\lim$ (if they exist) of the following sequences

(a) $(-1)^n/n$

(b) $(-1)^n(1 - 1/n)$

(c) $(-1)^n(1 + 1/n)^n$

(d) $\sum_{k=1}^n 1/k(k+1)$

(e) $n^n/n!$

2. Let a_n be a sequence of positive numbers such that $\lim a_n = L$. What can you say about the following limits?

(a) $\lim a_1 a_2 a_3 \dots a_n$

(b) $\lim (a_1 a_2 \dots a_n)^{1/n}$

3. Suppose $\{a_n\}$ is a sequence of positive numbers such that $\lim a_{n+1}/a_n = L$. Then what is $\lim a_n^{1/n}$?

4. What is $\lim n/(n!)^{1/n}$?

5. Use the L'Hopital's rule to establish that when x is small,

$$\ln(1+x) = x - x^2/2 + O(x^3).$$

6. Let $\mathcal{S}$ be the set of solutions of the differential equation

$$\partial u / \partial t = (1/2) \partial^2 u / \partial x^2.$$

Note that each solution $u(x, t)$ is a function of two variables. Is $\mathcal{S}$ a vector space?

References